



United International University (UIU)
Dept. of Computer Science and Engineering (CSE)
Final Exam: Summer 2025
Course Code: CSE 3811, Course Title: Artificial Intelligence
Total Marks: 40 Duration: 2 hours

Answer all questions. Marks are indicated on the right side of each question.

[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

1. A survey was taken among **60 Senior** chess players and **40 Young** chess players. They were asked which **Chess pieces** they liked the most. Based on the data, the following **CPT (Conditional Probability Table)** was found. [Note: “a” and “b” are two variables that have exact values]

Age (A)	Chess Piece (C)	P(C A)
Young	Queen	a
Young	Knight	3a
Senior	Queen	2b
Senior	Knight	3b

- a) Find out the values of **a** and **b**. [3]
b) What is the **percentage** of **Young** chess players who like the **Knight**? [3]

2. Harry, Ron and Hermione want to go to the next room. But the room is guarded by a dog named Fluffy. Now they need to put the dog to sleep in order to go to the next room. They have some options.

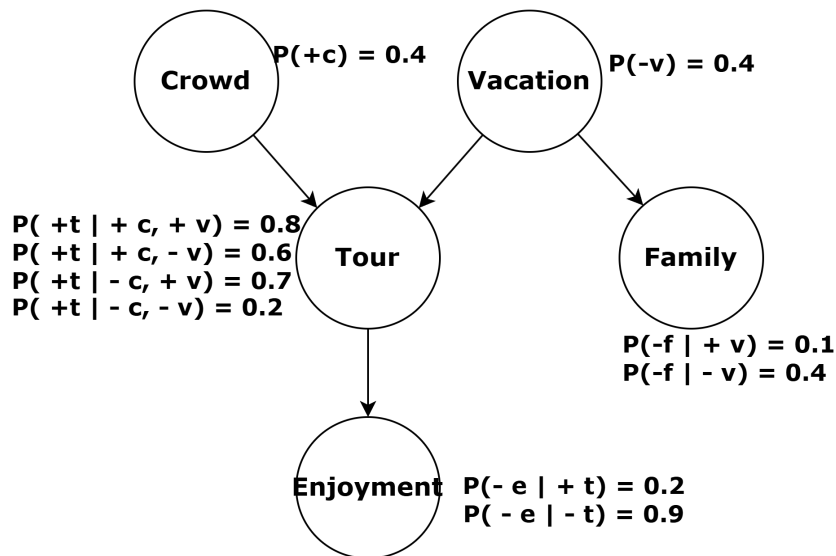
- They can use **Magic** to put the dog to **Sleep**. But using **Magic** will trigger the **Alarm**.
- They can use the **Harp** (a musical instrument) to put the dog to **Sleep**. But using **Harp** will make them **Tired**.
- If the dog is asleep, they can **Proceed** (move to the next room)

- a) Based on the instructions, build a Bayesian network. [2]
b) Answer **True or False** for the following statements [3]

[According to the network built in “(a)”]:

- i) $P(\text{Magic, Sleep} \mid \text{Harp}) = P(\text{Sleep} \mid \text{Harp})$
ii) $P(\text{Alarm, Proceed} \mid \text{Magic}) = P(\text{Alarm} \mid \text{Magic}) * P(\text{Proceed} \mid \text{Magic})$
iii) $P(\text{Proceed} \mid \text{Harp}) = P(\text{Proceed})$

- a) Given a new **Bayesian network below**, find out the **Probability of Enjoyment (+e)** given there is no crowd (-c). [5]



3. a) What is the **main assumption** made by the **Naïve Bayes classifier**? [1]
- b) The following dataset contains four features related to online shoppers: **Age Group**, **Income Level**, **Website Visit Frequency**, and **Purchase Amount**. The label, **Customer Type**, indicates whether a customer is Loyal, Occasional, or New. Use the **Naïve Bayes classifier** to determine the probability that a customer with the features **<Middle, High, Frequent, High>** belongs to the class Loyal. Show **detailed calculations and the final probability**. [7]

Customer Id	Age Group	Income Level	Visit Frequency	Purchase Amount	Customer Type
1	Young	High	Rare	High	Loyal
2	Middle	Medium	Rare	Low	New
3	Young	Low	Frequent	Medium	Loyal
4	Senior	High	Frequent	Low	New
5	Middle	High	Occasional	High	Loyal
6	Young	Medium	Occasional	Medium	Occasional
7	Middle	High	Frequent	High	Loyal
8	Senior	Low	Rare	High	New
9	Middle	High	Frequent	Medium	Occasional

4. You are given the opportunity to build a decision tree to predict the **performance rating** of employees based on the given attributes. Using the data below, determine which attribute is the **most important factor** in evaluating an employee's performance. [8]

Experience Level	Teamwork	Project Delivery	Performance Rating
High	Excellent	On Time	Excellent
Medium	Good	On Time	Good
Low	Poor	Late	Poor
High	Good	On Time	Excellent
High	Excellent	Late	Good
Medium	Poor	Late	Poor
Medium	Good	On Time	Good
Low	Good	Late	Poor
Medium	Excellent	On Time	Excellent
Low	Poor	Late	Poor
High	Good	On Time	Excellent
Medium	Good	On Time	Good

5. A music analytics company is studying listening patterns on **Spotify**, specifically tracking how users switch their primary artist preference among three major rap artists: **Eminem**, **Travis Scott**, and **Drake**. They collected data on users' most-played artists in **January 2023** and tracked how their preferences changed by **January 2024**.

- **Initial (January 2023) survey results (total of 180 users):**
 - **80** primarily listened to **Eminem**
 - **60** primarily listened to **Travis Scott**
 - **40** primarily listened to **Drake**
- One year later (**January 2024**), the following preference shifts were observed:
 - Among the **80 Eminem** listeners in **2023**:
 - **16** switched to **Travis Scott** as their primary artist
 - **12** switched to **Drake**
 - The rest stayed loyal to **Eminem**
 - Among the **60 Travis Scott** listeners in **2023**:
 - **15** switched to **Eminem**
 - **9** switched to **Drake**
 - The rest continued with **Travis Scott** as their top artist

- Among the **40 Drake** listeners in **2023**:
 - **10** switched to **Eminem**
 - **6** switched to **Travis Scott**
 - The rest kept **Drake** as their most-played artist

Assume these **transition patterns** remain the same for each subsequent year (**2024 → 2025**, **2025 → 2026**, **2026 → 2027** etc.).

Questions

1. Construct a **transition matrix** representing the listener preference changes from any year **t** to the next year **t+1**. **[2]**
2. Predict the percentage of users who will have **Travis Scott** as their primary artist in **2028** using the initial survey of **2023**. **[3]**
3. Calculate the **long-run probabilities** that a randomly selected Spotify user will have for each of the three artists. **[3]**